

Khoshaba Odeesho

Data Scientist | MLOps Engineer | AI Automation Engineer
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[Live Dashboard: prop-market-intelligence-khoshaba-odeesho.streamlit.app](https://prop-market-intelligence-khoshaba-odeesho.streamlit.app)

PROFESSIONAL SUMMARY

Self-taught Data Scientist and AI Automation Engineer based in Melbourne, with 25+ public GitHub projects covering machine learning, MLOps, NLP, business intelligence, and AI agent workflows. I build and ship end-to-end ML pipelines using Python, XGBoost, Docker, and FastAPI — and design autonomous n8n automation systems integrated with LLM APIs (Groq, Gemini, OpenRouter) that remove manual work from real business processes. My background combines 13 years of hotel operations management with hands-on technical depth, giving me a practical, business-first approach to data science. Oracle Hospitality certified (Opera Cloud PMS). IBM AI Fundamentals certified. AWS Cloud Practitioner certified. CSP Visa granted — full working rights in Australia.

TECHNICAL SKILLS

Languages & Data: Python (pandas, NumPy, scikit-learn, XGBoost), SQL (window functions, CTEs, subqueries, joins), Jupyter Notebook

Machine Learning: Classification, regression, Optuna hyperparameter tuning, SMOTE, SHAP explainability, MLflow experiment tracking, NLP, LSTM

MLOps & Cloud: Docker, FastAPI, REST APIs, GitHub Actions CI/CD, AWS Sydney (EC2, S3), Apache Airflow, Evidently AI, Prometheus, Grafana

AI & Automation: n8n (workflow and agent design), RAG, Groq API, Gemini API, OpenRouter, Pinecone vector store, LLM prompt engineering

BI & Visualisation: Power BI (DAX, data modelling, 10+ measures), Tableau, Streamlit, Plotly, Dash, Matplotlib, Seaborn

Database: Supabase PostgreSQL (schema design, indexing, views, parameterised queries), SQL aggregations and analytics

Web Development: TypeScript, Next.js, React, Vercel — deployed live SaaS applications

Hospitality Technology: Opera Cloud PMS (Oracle Hospitality Certified 2026), Opera On-Premise, Protel PMS, GPMS, Omega PMS (Trainer)

Excel & Office: Advanced Excel (VBA macros, Pivot Tables, Power Query, VLOOKUP, dashboard automation), Microsoft Office Suite

KEY PROJECTS

Melbourne Northern Corridor — Property Intelligence Platform | 2025–2026

Live: prop-market-intelligence-khoshaba-odeesho.streamlit.app | github.com/Assyrian91/prop-market-intelligence | Python · PostgreSQL · XGBoost · Gemini AI · Streamlit · Power BI · n8n · Supabase

- Designed and deployed a fully automated, end-to-end property intelligence platform covering 7 Melbourne northern corridor suburbs across 12 years of real Victorian Government data — resulting in a live public dashboard that any employer or client can visit and interact with today.
- Built a 6-table PostgreSQL schema on Supabase using window functions, LAG-based year-on-year calculations, CAGR formulas, suburb ranking, and a permanent analytical view — demonstrating production-grade SQL data engineering skills directly applicable to analyst and analytics engineer roles.
- Trained XGBoost and Linear Regression price prediction models on 55 historical rows, achieving an XGBoost MAE of \$11,744 (1.8% average error) on real 2024 holdout data — with best single prediction 0.42% off actual price — proving that carefully engineered features outperform data volume.
- Integrated Google Gemini 2.0 Flash via API to automatically generate 8 suburb-level market reports and a corridor executive summary from live database data, eliminating manual report writing entirely.
- Built a 5-page Streamlit dashboard (publicly deployed on Streamlit Cloud) and a 4-page Power BI report with 10 custom DAX measures, both reading live from Supabase PostgreSQL — demonstrating fluency across both the Python BI stack and the Microsoft BI stack simultaneously.
- Automated the full daily pipeline with a 10-node n8n workflow that triggers at 7am, queries Supabase, builds a structured AI prompt, calls OpenRouter, saves the generated report to the database, and delivers a formatted briefing to Telegram — zero manual work required after deployment.

Australian Hotel Booking Cancellation Prediction — End-to-End MLOps | 2025–2026

github.com/Assyrian91/au-hotels-cancellation | Python · XGBoost · FastAPI · Docker · Apache Airflow · AWS Sydney

- Achieved AUC-ROC of 0.874 and 80.1% accuracy on 87,000+ real Australian hotel bookings by engineering 13 Australia-specific features — including state school holidays, ANZAC Day, AFL Grand Final, Melbourne F1 Grand Prix, and AUD price tiers — that generic international datasets overlook entirely.
- Eliminated manual model maintenance by building an Apache Airflow DAG that automatically retrains the model every week whenever AUC drops below 0.85 or the dataset grows by 10%, keeping predictions accurate without any human intervention.

- Made the system genuinely production-ready by containerising a FastAPI REST service in Docker, deploying it to AWS Sydney, wiring GitHub Actions CI/CD for automatic testing and deployment, and adding Evidently AI drift detection with Prometheus metrics and a Grafana live health dashboard.

Guest Agent — Live AI Hotel Concierge (n8n + RAG) | 2025–2026

github.com/Assyrian91/guest-agent-live-workflow | n8n · Groq API · RAG · Pinecone · Telegram Bot

- Delivered 24/7 AI guest support for La Roche Hotel by building a conversational concierge in n8n that retrieves answers exclusively from real hotel policy documents using RAG — eliminating an entire category of front-desk queries without requiring any additional staff.
- Prevented hallucinated responses by grounding every AI answer in retrieved document text rather than model memory, with automatic escalation to a human agent whenever the system's confidence falls below a defined threshold.

Assyrian AI Automation — Universal Workflow Mapper (Live SaaS) | 2025–2026

ai-workflows-eta.vercel.app | github.com/Assyrian91/Ai-workflows | n8n · Groq · React/Next.js · Vercel

- Enabled business owners across 8+ industries — law, medical, hospitality, e-commerce, recruitment — to identify their highest-ROI automation opportunities by building a live SaaS product that takes a business description and returns a ranked automation roadmap with time and cost estimates.
- Shipped the complete product solo — from n8n automation logic and Groq AI processing to a React/Next.js frontend deployed on Vercel — without a team, existing codebase, or prior web development background.

MLOps Customer Churn Pipeline | 2025

github.com/Assyrian91/mlops-churn-pipeline | Python · Docker · FastAPI · scikit-learn · MLflow

- Replaced a one-off analysis notebook with a fully reusable, production-style MLOps pipeline — covering data ingestion, feature engineering, model training, experiment tracking with MLflow, and REST API model serving — containerised with Docker so it deploys consistently across any environment.

Credit Card Fraud Detection System | 2025

github.com/Assyrian91/fraud-detection-project | Python · XGBoost · SMOTE · FastAPI

- Significantly improved fraud-class recall on a severely imbalanced dataset by applying SMOTE oversampling and class-weight tuning, then systematically compared Logistic Regression, Random Forest, and XGBoost before deploying the strongest model through a scikit-learn pipeline to a FastAPI endpoint.

ADDITIONAL PROJECTS

- Assyrian AI Suite (12 live tools) — Streamlit and Groq-powered business toolkit including hotel concierge bot, restaurant assistant, dental clinic assistant, document summariser, and content writers — deployed publicly at ashuraya-ai.streamlit.app
- Personal Automation Assistant — Autonomous n8n AI agent that connects to multiple personal data sources and uses working memory across a conversation to decide which tool to call — a practical demonstration of agentic AI design.
- AI Job Market Analysis — Scraped and analysed 15,000+ AI job postings using NLP-based skill extraction; built an interactive dashboard of salary distributions and skill-demand heatmaps across roles and locations.
- Full portfolio also includes: Dynamic Pricing (Uber-style simulation with Dash dashboard), Smart Loan Recovery, Movie Sentiment Analysis (LSTM), Car Price Prediction, Power BI Uber Dashboard, Tableau Premier League Dashboard — github.com/Assyrian91

PROFESSIONAL EXPERIENCE

Note: All roles below are full-time hotel management positions. Bullets highlight data, reporting, and process-improvement contributions relevant to a data science career. Full hospitality CV available on request.

Reception Manager / Duty Manager / Operations Manager | Sep 2021 – Apr 2024

La Roche Hotel — 4-Star | Opera PMS, Protel PMS | Erbil, Iraq

- Reduced guest complaints and improved service quality scores by systematically analysing 1,000+ feedback records and presenting specific, evidence-based improvement recommendations directly to the General Manager — an early example of using data to drive operational decisions.
- Improved rostering accuracy and eliminated scheduling conflicts for a 120+ person team by building shift schedules grounded in occupancy forecasts and demand patterns extracted from Opera PMS reports.
- Maintained accurate revenue reporting across 1,000+ annual events by designing and enforcing data-flow coordination between front office, F&B, housekeeping, and security — a process that is now a model for the property.

Omega PMS Trainer (Contract) | 2022 – 2024

Freelance | Erbil, Iraq

- Reduced front-office processing errors across multiple hotel properties by designing structured, hands-on Omega PMS training programmes and tailored SOPs covering reservations, night audit, and revenue reporting — delivered directly to staff at each property.

Duty Manager | Jun 2018 – Sep 2020

Dilshad Palace Hotel — 5-Star | GPMS | Duhok, Iraq

- Reduced labour costs by approximately 15% while maintaining service standards and team morale by redesigning shift scheduling using 12 months of GPMS historical demand data — an early, real-world application of data-informed operational optimisation.

Night Shift Manager / Night Auditor | Feb 2015 – Jun 2018

Dilshad Palace Hotel — 5-Star | GPMS | Duhok, Iraq

- Maintained consistently high financial accuracy across years of nightly revenue reconciliation across rooms, F&B, and ancillary revenue centres, and built lasting data-entry discipline by training 100+ staff on correct GPMS procedures.

Receptionist | Nov 2011 – Aug 2014

Hakar Hotel — 4-Star | Opera PMS | Duhok, Iraq

- Built the foundational PMS proficiency and financial-accuracy habits that underpinned a 13-year hotel career through daily cash reconciliation and reservation data entry in Opera PMS across three years of front-desk operations.

EDUCATION

Bachelor of Mathematics Science — University of Duhok, Iraq (final year, 2 subjects remaining)

CERTIFICATIONS

- Oracle Hospitality — Opera Cloud PMS Certification (Completed 2026)
- IBM Artificial Intelligence Fundamentals — IBM SkillsBuild, Jul 2025 | Verify: credly.com/badges/dc8beb8b-bd3a-48aa-b9f8-37c70f846b26
- Data Science Essentials with Python — Cisco Networking Academy, Jul 2025
- AWS Cloud Practitioner Certification — Amazon Web Services
- Google Data Analytics — Coursera
- Data Analytics Bootcamp — Alex Freberg (Data with Alex)
- Data Analytics Essentials — Cisco Networking Academy, Jul 2024
- Introduction to Data Science — Cisco Networking Academy, Jun 2024
- Introduction to Cybersecurity — Cisco Networking Academy, Jun 2024
- Microsoft Excel for Business (Advanced Excel, VBA, Automation) — Coursera
- Data Analysis, MicroWork and English — WorkWell / GIZ German Cooperation / Duhok Tech Hub (88 hours)

ADDITIONAL INFORMATION

Work Rights: CSP Visa Granted — full working rights in Australia

Languages: English (Professional Working Proficiency) | Arabic (Native) | Assyrian (Native)

References available on request